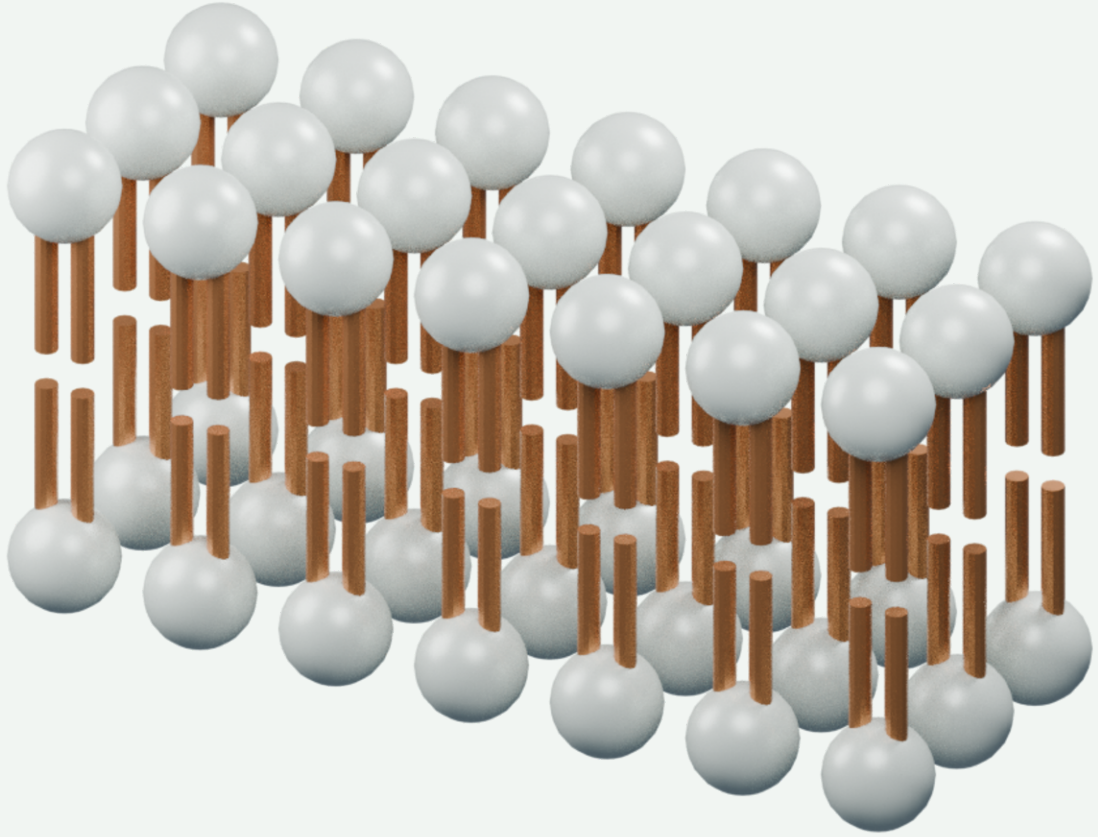
A boundary.
Not a barrier.

The cell membrane regulates exchange between a cell and its surroundings.



Molecular schematic; not to scale

Diffusion follows a concentration gradient.

01

Particles move randomly.

02

More particles leave the crowded side.

03

Net movement declines toward equilibrium.

Not every molecule crosses the same way.

Simple diffusion

Small nonpolar molecules can pass through the lipid bilayer.

Facilitated diffusion

Transport proteins help selected solutes move down their gradients.

Against the gradient? Energy is required.

Active transport couples movement to an energy source. It is not simply faster diffusion.

Ask what drives the movement.

Identify the solute, the membrane pathway and the direction of the concentration gradient.